

$$\underbrace{\frac{\partial L}{\partial \beta}}_{m \times 1} = -\frac{2}{n} \underbrace{X^T}_{m \times n} \underbrace{(y - \hat{y})}_{n \times 1}$$

one row per coefficient:

for coefficient j , the errors are multiplied by column j of X , summed, and divided by n